Entangled Meanings: Classification and Ambiguity Resolution in Near–Term QNLP

Chi Zhang
NLP-Lab
Department of Mathematics
Indiana University
Bloomington, IN, USA
czh4@iu.edu

Akriti Kumari
NLP-Lab
Luddy School of Informatics, Computing, and Engineering
Indiana University
Bloomington, IN, USA
akkumari@iu.edu

Damir Cavar
NLP-Lab
Indiana University
Bloomington, IN, USA
dcavar@iu.edu

2024 IEEE International Conference on Quantum Computing and Engineering (QCE) | 979-8-3315-4137-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/QCE60285.2024.10355

***Abstract*—We demonstrate two tasks in Quantum Natural Language Processing (QNLP): classification and disambiguation.**

For the classification task, we utilized an amplitude encoding algorithm and achieved perfect accuracy on the standard `lambeq` dataset, which is commonly used for benchmarking in quantum NLP. Additionally, we obtained accuracy from 55% to 72.5% on a more complex and realistic Amazon review dataset, which is still a reasonable result, given the current capability of quantum computing and QNLP. Additionally, we found that UMAP is the best dimension reduction method in our experiment setting. All classification results were done on the `default.qubit` simulator in `pennylane 0.36` python library. Our classification results highlight the potential of quantum algorithms in practical applications.

For the disambiguation task, we selected 18 ambiguous nouns, 32 unambiguous nouns, and 18 different verbs. Our experiments using the `QASM simulator` within the `qiskit` Python library demonstrated that the simulator could perfectly differentiate between the various meanings of ambiguous nouns in different contexts, achieving perfect differentiation. Furthermore, we extended our study to a real quantum device, the `ibm_kyoto` quantum computer. There, we tested our disambiguation approach on a subset of 4 random nouns (two ambiguous and two unambiguous) and observed that `ibm_kyoto` could achieve an accuracy range of 82.1% to 98.9% in disambiguation tasks, extending the datasets and improving the results of existing ambiguity resolution experiments [1].

Our work demonstrates the capability of quantum computing in dealing with real-world NLP tasks, hence contributing to the advancement of both QML and NLP.

***Index Terms*—Natural Language Processing, Semantics, Sentiment Analysis, Supervised Machine Learning**

I. Introduction

Quantum computing algorithms have advanced significantly since the 1990s [2], [3], with recent practical quantum computers accelerating progress. Near-term quantum computers have shown achievements in chemistry [4], finance [5], and machine learning [6].

Quantum Natural Language Processing (QNLP), a subarea of Quantum Machine Learning (QML) [7], aims to perform Natural Language Processing (NLP) on quantum computers. Two main approaches exist one based on mathematical frameworks like pre-group grammar [8] and category theory and the other on machine learning algorithms. The first school, including authors like Bob Coecke, established foundational frameworks integrating vector space models and pre-group grammar [9], [10], and conducted the first practical NLP experiment on quantum computers in 2022 [11]. The second school, including authors like Widdows, performed classification tasks on realistic IMDB datasets [12] and the first QNLP ambiguity resolution task on an 11-qubit IonQ quantum computer [1].

This proposal demonstrates the capability of quantum computers in classification and ambiguity resolution tasks on real-life datasets, improving upon the algorithms and methodologies from prior work [1].

II. Approaches

A. Classification

We performed binary classification on the `lambeq` [13] and Amazon review [14] datasets. We converted the texts using `Word2Vec` in `gensim 4.1.2` [15]and `spaCy 3.7` [16]. We then applied PCA, LDA, t-SNE from `scikit-learn` [17], and UMAP from `umap-learn` [18] to the word vectors. Next, we used amplitude and divide-and-conquer encoding algorithms from [19] with the `qml` module from `pennylane 0.37.0` [20].

We trained our pipelines on the default.qubit simulator from pennylane for 15 iterations with 150 steps each and a learning rate of 0.1, using a 6:2:2 train-validation-test ratio. We listed the training, validation, and test accuracy of all methods to find the best combination for limited computing resources. The code can be found in the GitHub repository.

B. Ambiguity Resolution

Language is inherently ambiguous, and differentiating this ambiguity in context is crucial [21]. Early researchers used matrices and vectors to analyze texts [22], [23]. Authors like Widdows [1] represented nouns as vectors and transitive verbs as matrices.

Inspired by these authors, we also treat nouns as vectors and transitive verbs as matrices. We classify nouns into 16 categories and verb phrases into 10 categories. Verbs are viewed as functions that take nouns and return verb phrases, represented as 10×16 matrices. These classifications can be found in our GitHub repository.

979-8-3315-4137-8/24/$31.00 ©2024 IEEE
DOI 10.1109/QCE60285.2024.10355

Authorized licensed use limited to: Universita degli Studi di Salerno. Downloaded on July 02,2026 at 10:12:32 UTC from IEEE Xplore. Restrictions apply.

We then convert the vectors and matrices into quantum circuits using X and CX gates because the qubits are at default value $|0\rangle$ and become $|1\rangle$ after X gates are applied in most quantum computing libraries. Examples of the quantum circuits for *Fan*, *Admire*, and *Admire Fan* are shown in the poster.

III. Data and Results

A. Classification

The `lambeq` dataset and the Amazon review dataset can be found in our GitHub repository.

We listed classification results with test accuracy over 53% in Table 2 in the poster, highlighting those above 80%.

We achieved perfect test accuracy on the `lambeq` dataset using only 1 qubit, compared to 5 qubits in [11] and 4 qubits in [1]. UMAP proved to be the most economical and accurate dimension reduction method.

Our test accuracy on the Amazon review language ranged from 55% to 72.5%, comparable to the 57% to 62% accuracy [1] on the IMDB dataset [12].

B. Ambiguity Resolution

Our dataset includes 18 ambiguous nouns (*e.g., Fan, Mole, Date, Jam*), 32 unambiguous nouns (*e.g., Screwdriver, Infiltration, Dolphin, Book*), and 18 transitive verbs (*e.g., Examine, Emphasize, Catch, Cultivate*). The whole corpus can be found in the GitHub repository.

We tested ambiguity resolution on `qasm_simulator` from `qiskit-aer 0.14.2` for all nouns. Due to limited computing time, we tested only 4 randomly chosen nouns (2 ambiguous and 2 unambiguous) on `ibm_kyoto`. On `qasm_simulator`, the accuracy for ambiguity resolution was 100% for all nouns, while on `ibm_kyoto`, the accuracy ranged from 82.1% to 98.9%. The results for the 4 randomly chosen words are in Table 2 in the poster.

Data Availability

All code and data discussed in this work are available in the GitHub repository https://github.com/chizhang24/entangled-meanings. The models used are all freely available online.

Acknowledgment

The authors would like to express sincere gratitude to Larry Moss for his invaluable guidance and Dhananjay Srivastava for helpful discussions.

References

[1] D. Widdows, A. Alexander, D. Zhu, C. Zimmerman, and A. Majumder, "Near-Term Advances in Quantum Natural Language Processing," *Annals of Mathematics and Artificial Intelligence*, Apr. 2024.

[2] P. W. Shor, "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer," *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, Oct. 1997, publisher: Society for Industrial and Applied Mathematics.

[3] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing*, ser. STOC '96. New York, NY, USA: Association for Computing Machinery, Jul. 1996, pp. 212–219.

[4] B. Bauer, S. Bravyi, M. Motta, and G. K.-L. Chan, "Quantum Algorithms for Quantum Chemistry and Quantum Materials Science," *Chemical Reviews*, vol. 120, no. 22, pp. 12 685–12 717, Nov. 2020.

[5] D. Emmanoulopoulos and S. Dimoska, "Quantum Machine Learning in Finance: Time Series Forecasting," Feb. 2022.

[6] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, Sep. 2017.

[7] D. Peral-García, J. Cruz-Benito, and F. J. García-Peñalvo, "Comparing Natural Language Processing and Quantum Natural Processing approaches in text classification tasks," *Expert Systems with Applications*, vol. 254, p. 124427, Nov. 2024.

[8] J. Lambek, *From Word to Sentence: A Computational Algebraic Approach to Grammar*. Polimetrica s.a.s., 2008.

[9] B. Coecke, M. Sadrzadeh, and S. Clark, "Mathematical Foundations for a Compositional Distributional Model of Meaning," *Lambek Festschirft, special issue of Linguistic Analysis, 2010.*, vol. abs/1003.4394, 2010. [Online]. Available: https://ora.ox.ac.uk/objects/uuid:059c2872-ab48-4c27-b52e-018873b04460

[10] B. Coecke, "The Mathematics of Text Structure," Feb. 2020.

[11] R. Lorenz, A. Pearson, K. Meichanetzidis, D. Kartsaklis, and B. Coecke, "QNLP in Practice: Running Compositional Models of Meaning on a Quantum Computer," *Journal of Artificial Intelligence Research*, vol. 76, pp. 1305–1342, Apr. 2023.

[12] A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning Word Vectors for Sentiment Analysis," in *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, D. Lin, Y. Matsumoto, and R. Mihalcea, Eds. Portland, Oregon, USA: Association for Computational Linguistics, Jun. 2011, pp. 142–150.

[13] D. Kartsaklis, I. Fan, R. Yeung, A. Pearson, R. Lorenz, A. Toumi, G. de Felice, K. Meichanetzidis, S. Clark, and B. Coecke, "Lambeq: An Efficient High-Level Python Library for Quantum NLP," Oct. 2021.

[14] D. Kotzias, "Sentiment Labelled Sentences," 2015.

[15] R. Rehurek and P. Sojka, "Gensim–python framework for vector space modelling," *NLP Centre, Faculty of Informatics, Masaryk University, Brno, Czech Republic*, vol. 3, no. 2, 2011.

[16] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, "spaCy: Industrial-strength Natural Language Processing in Python," 2020.

[17] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[18] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," Sep. 2020.

[19] I. F. Araujo, D. K. Park, F. Petruccione, and A. J. Da Silva, "A divide-and-conquer algorithm for quantum state preparation," *Scientific Reports*, vol. 11, no. 1, p. 6329, Mar. 2021.

[20] V. Bergholm *et al.*, "PennyLane: Automatic differentiation of hybrid quantum-classical computations," Jul. 2022.

[21] J. Aitchison, *Words in the Mind: An Introduction to the Mental Lexicon, 3rd Ed*, ser. Words in the Mind: An Introduction to the Mental Lexicon, 3rd Ed. Malden: Blackwell Publishing, 2003.

[22] H. Schütze, "Automatic Word Sense Discrimination," *Computational Linguistics*, vol. 24, no. 1, pp. 97–123, 1998, place: Cambridge, MA Publisher: MIT Press. [Online]. Available: https://aclanthology.org/J98-1004

[23] M. Baroni and R. Zamparelli, "Nouns are Vectors, Adjectives are Matrices: Representing Adjective-Noun Constructions in Semantic Space," in *Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing*, H. Li and L. Màrquez, Eds. Cambridge, MA: Association for Computational Linguistics, Oct. 2010, pp. 1183–1193.

Authorized licensed use limited to: Universita degli Studi di Salerno. Downloaded on July 02,2026 at 10:12:32 UTC from IEEE Xplore. Restrictions apply.